SW3000R TSN REDBOX

SUPPORT： 17265076798

TSN SWITCH

SW3000R TSN REDBOX

FEATURES:

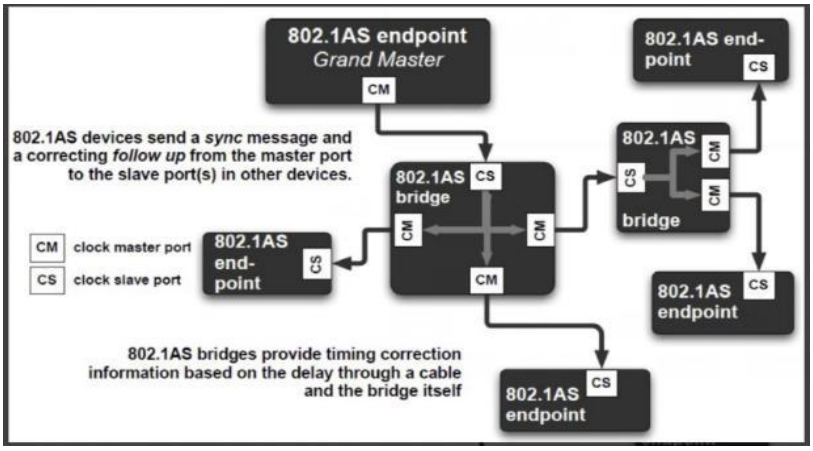
- 25 Ports 100/1000Base-T
- 4 Ports1/2.5/10GBase-T SFP
- 1 PCIe Endpiont
- gPTP BC and SyncE
- TSN (802.1Qav/bv/at/ci/bu)
- 802.1AS, IEEE1588/PTP
- 1PPS/ToD,10Mhz OUT
- TCXO and DPLL phase-locked
- HSR/PRP (Redbox)
- Web and CLI config
- Optional PoE

SW3000R TSN REDBOX SWITCH
USER MANUAL
100/1000BASE-T AND 10G SFP+ TSN SWITCH

SW3000R TSN Redbox SWITCH

The 30-port high-performance TSN (Time-Sensitive Networking) switch is equipped with 25 channels of Gigabit adaptive RJ45 interfaces (with optional PoE support), 4 channels of 1/2.5/5/10G multi-speed SFP+ interfaces, and one PCIe-endpoint interface. It supports SyncE frequency synchronization, IEEE 1588-PTP, and 802.1AS-gPTP. It also provides 1PPS/ToD input/output and a 10MHz atomic clock output. Supports L2 switching and L3 routing, along with complex management features such as 802.1Q, QoS, and port control. Suitable for automotive/aeronautical gateway and AVB implementations. Additionally, it supports gPTP Boundary Clock (BC) functionality, a full suite of TSN protocols, and HSR/PRP (RedBox) link redundancy mechanisms.

Architecture：

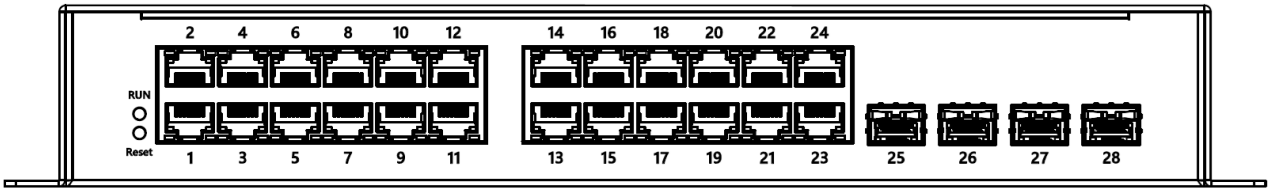


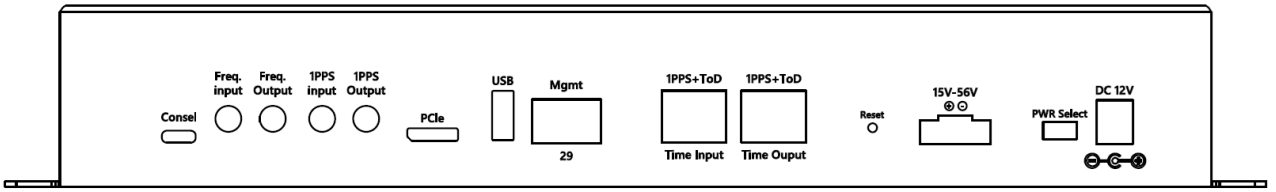
FEATURES：

- Wide voltage range: 8V~56V,<28W, Operating temp: -40℃ to +85℃
- Clock synchronization systems for eVTOLs, vehicles, as well as testing for Lidar, ADAS, and other applications.Used in industrial control and aerospace.
- Complete TSN functionality, TSN/AVB protocol(802.1Qav/bv/at/ci/bu)
- Supports HSR/PRP (Redbox)
- With overvoltage protection, reverse polarity protection, and surge protection, suitable for long-term operation
- Dimensions: 305 x 204 x 48 mm

1CONFIG

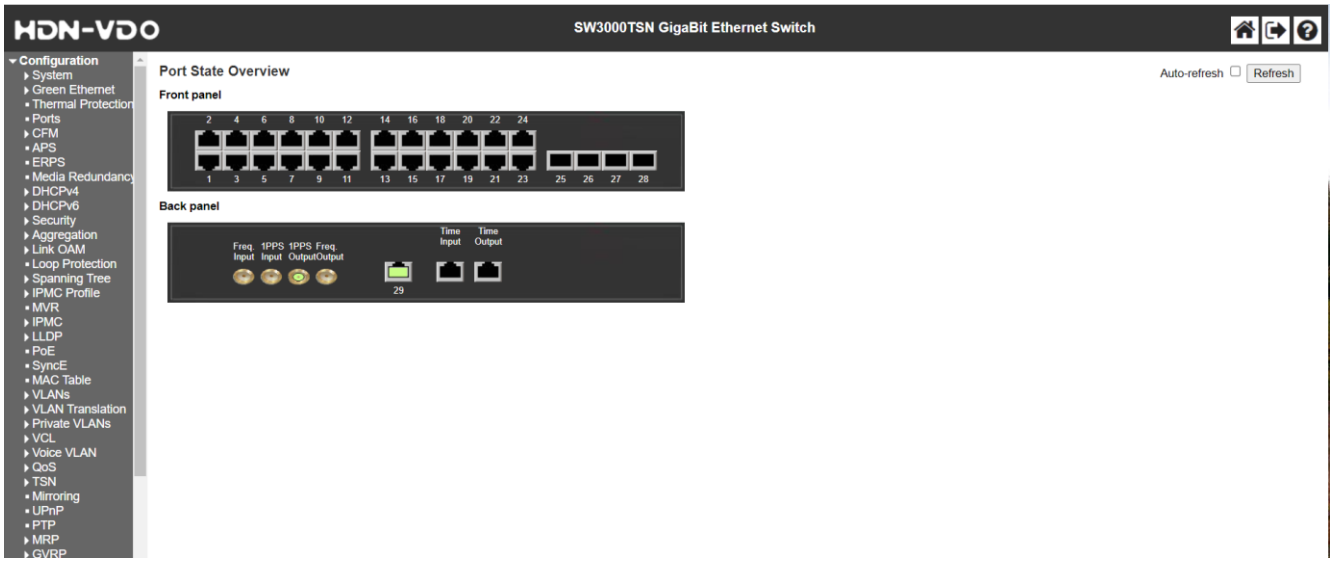
The SW3000R TSN series switches possess a variety of configuration functions, enabling them to provide customers with robust application capabilities and meet their more precise application needs. The following is a demonstration of the SW3000R TSN interfaces:



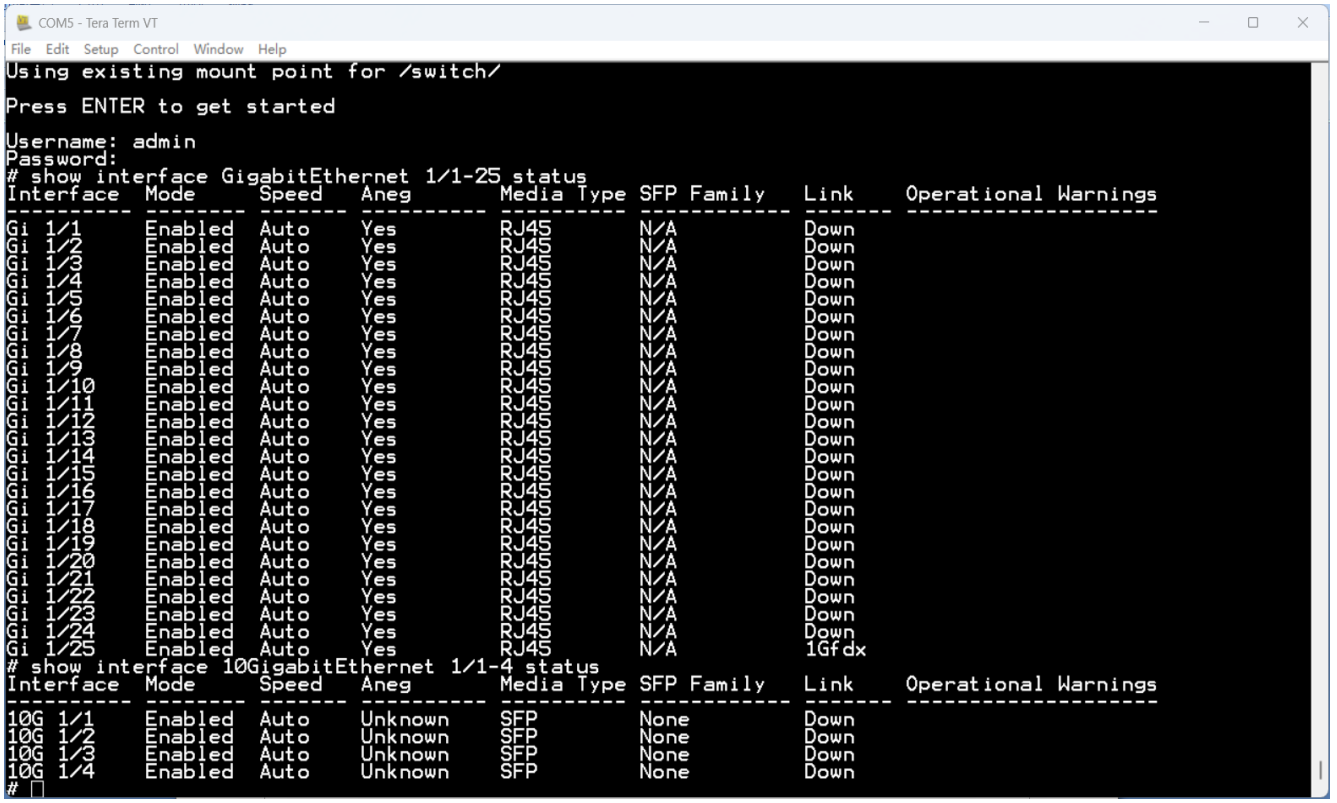


This product supports dual management methods, Web and CLI configuration, enabling customers to achieve flexible configuration needs for all functions.

Web Configuration Method: By default, the product uses Port 29 as the configuration port, with management IP address 192.168.1.10. Default username is "admin" and the password is initially empty. Users can log in and set their own password. Below is the switch configuration interface:



CLI Configuration Method: The product utilizes a USB type C port for CLI command configuration. Customers are required to install the USB serial port driver in advance (please contact after-sales technical support for the driver package). Using Tera Term, you can manage and configure the switch. The account name is "admin" and the password is initially empty. Users can log in and set their own password.



2 SOLUTIONS

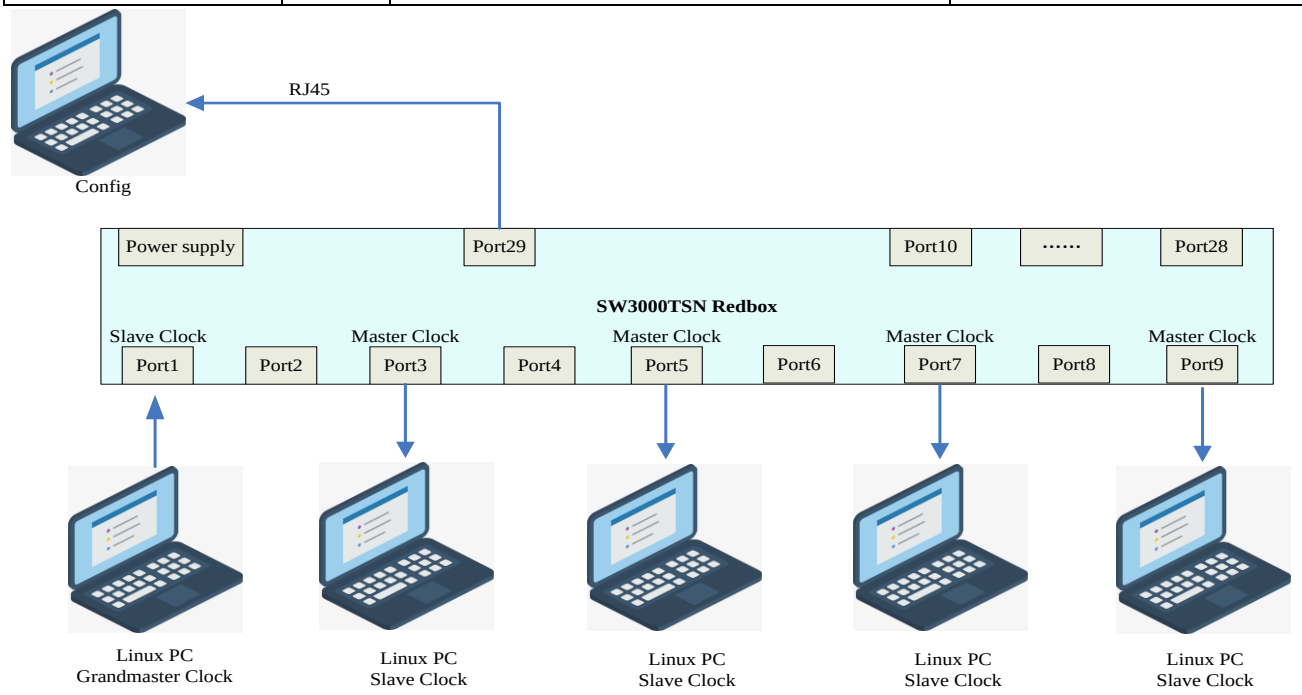
2.1 Dual-channel collects communication data between two devices (Bypass function)

This product can capture communication messages between DUT1 and DUT2, and can also simulate nodes to send messages to DUT1 and DUT2. Additionally, it can facilitate data exchange between multiple master devices and multiple slave devices. Communication data of each master node can be viewed through monitoring software.

2.2 Building a gPTP clock bridge

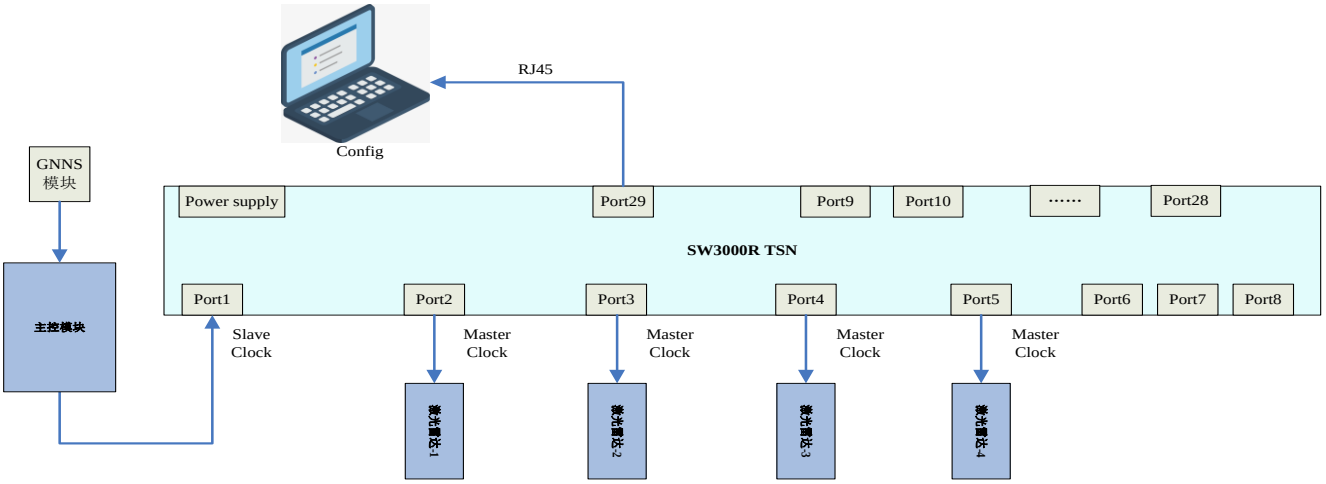
The TSN test network aims to establish the essential network architecture required for precise time synchronization and for simulating the precise time distribution within vehicle internal controller modules. It necessitates one grandmaster, one or multiple slave clock, one TSN switch, and one configuration computer. Among these, the grandmaster and slave clock utilize Linux computers (with network cards supporting IEEE1588). The system architecture is outlined as follows:

Name	Amount	Functionality	Comment
TSN switch	1	Built-in TSN protocol stack, boundary clock timing	SW3000R TSN
LinuxPC (IEEE1588)	5	Used for simulating gPTP clock nodes	The number of nodes can be reduced according to specific needs
Win/linux PC for config	1	Monitoring data and configuring functions	
RJ45 Ethernet Cable	1	Used to connect computer and TSN switch	



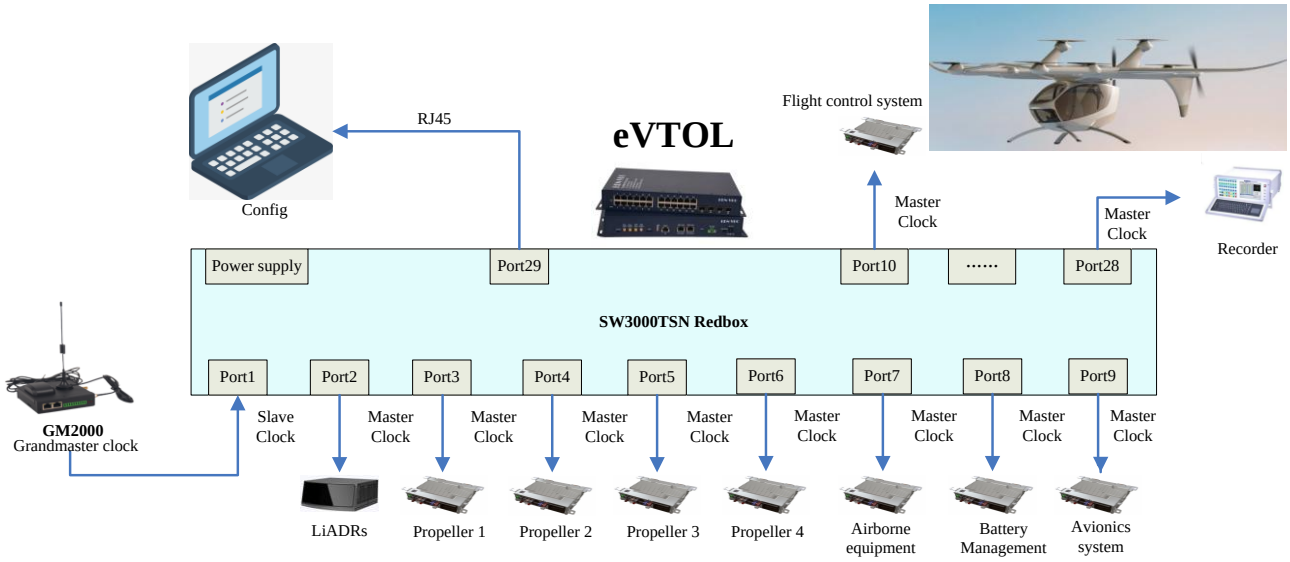
2.2 gPTP clock bridge case for automotive LiDAR

A simplified Lidar bridging scheme is established as follows, which supports gPTP time synchronization and achieves data transmission and precise synchronization between the main control module and the Lidar.



2.3 eVTOL system solution

A simplified eVTOL (Electric Vertical Take-Off and Landing) system solution that supports gPTP. This solution is applicable for time synchronization among multiple subsystems in eVTOL systems, including radar, avionics equipment, flight control systems, power systems, battery management systems, data recording systems, high-definition cameras, and other devices requiring synchronization.



3 BRIEF INSTRUCTIONS FOR USE

3.1 QUICK START

Connect one channel of the SW3000R TSN Ethernet switch to the corresponding test target. Supply the 12V power to the switch. At this time, the yellow Init light flash one time and green RUN light will light up once the Switch boot completely. When the green light on, it means that the switch initialization is complete and can be used. At this point, use a regular Ethernet cable to connect the computer to the Port29.

Definition and Explanation of LED Lights:

LED name	color	Function Indication
Reset LED	yellow	Light up after startup or restart one time
RUN LED	green	Solid green after normal run.
Channel LED	yellow /green	Light up shows link up. Flashing when there is data communication.

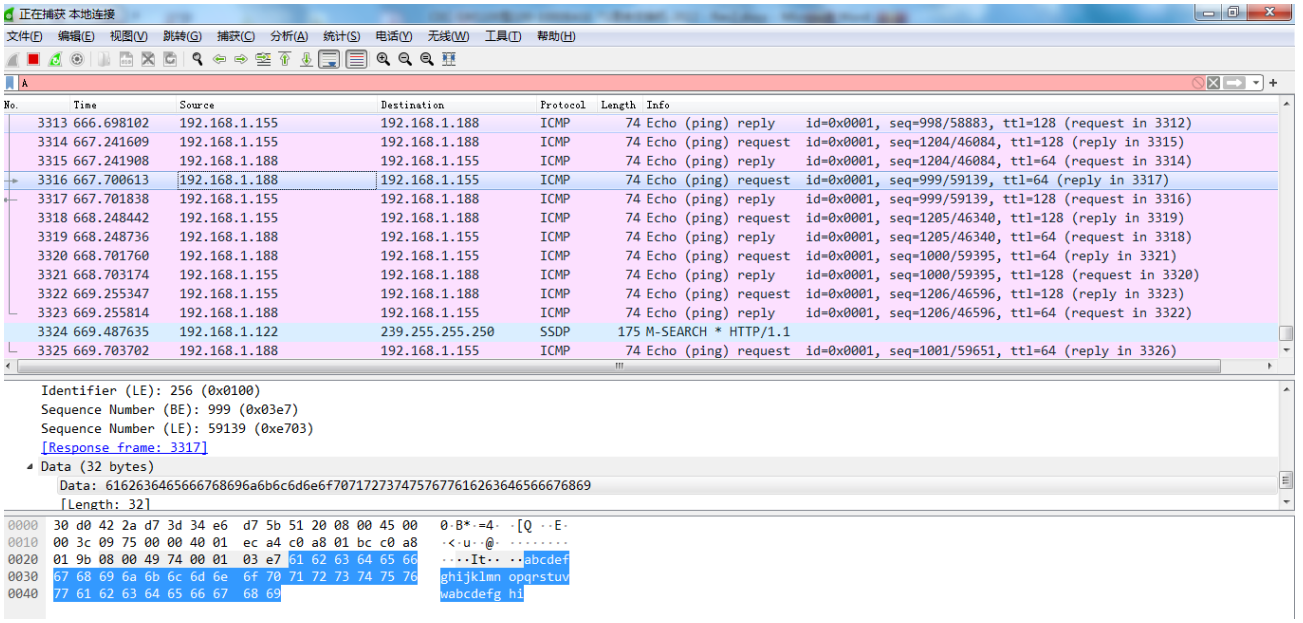
For example, IP address of the DUT is 192.168.1.122, Configure computer's IP address to 192.168.1.155, which is in the same LAN segment. In this way, you can open the DOS terminal on your computer and use the Ping 192.168.1.122 -t command to send data packets to the target device.

Test case:

DUT1 (IP: 192.168.1.122), and DUT2 (IP: 192.168.1.188). Connect the two devices to CH1 and CH2 of SW3000R TSN respectively Among them, The indicator lights corresponding to channels CH1 and CH2 are working, indicating that a connection has been established and the indicator lights will flash when there is data communication

Then, connect the SW3000R TSN to the computer using an Ethernet cable and configure the computer's IP to be on the same network segment as the two devices (IP: 192.168.1.155), Open Wireshark software to view the communication data between DUT1 and DUT2

WiresharkDisplay as follows:



Of course, you can also use the data collection software and Ethernet transceiver testing tools we provide:

Wireshark Tool:

https://wwm.lanzouh.com/iK3eQtjvfti Password:d5te

Data send tool:

https://wwm.lanzouh.com/i3bNTtjv7hi Password:4pd7

Testing tool:

https://wwm.lanzouo.com/b01psw5vi Password:bn3z

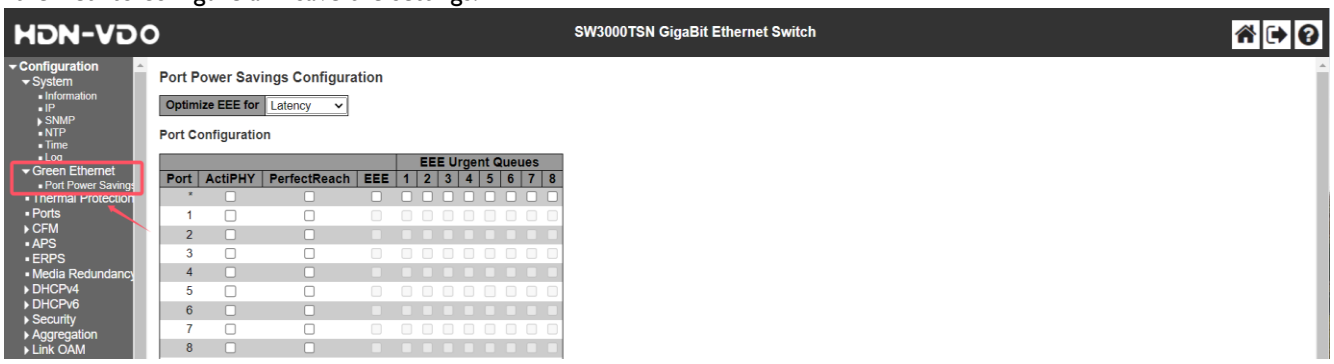
3.2 EXPLANATION OF FUNCTIONS

3.2.1 SYSTEM INFORMATION CONFIGURATION

This system enables customers to set up customized customer information. Simply click on "Configuration" → "System" → "Information" to make the settings and save them.

3.2.2 GREEN ETHERNET

This system allows customers to enable the port power-saving function. Simply click on "Configuration" → "Green Ethernet" to configure and save the settings.



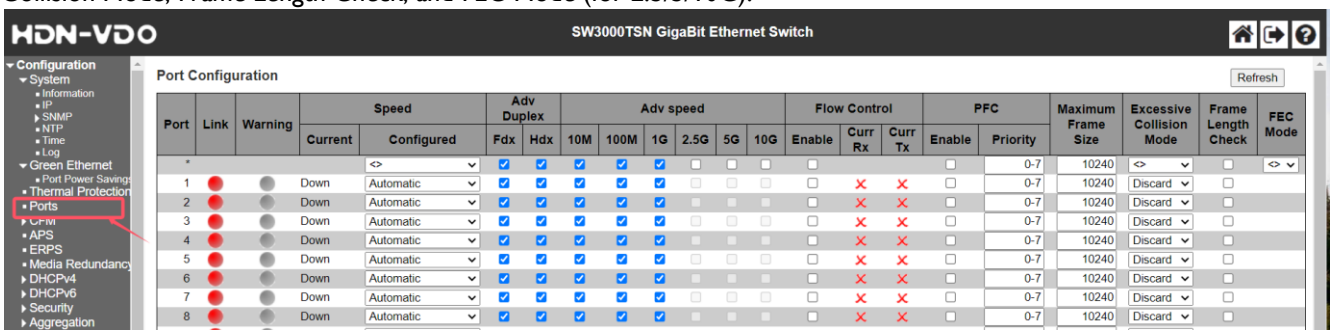
3.2.3 THERMAL PROTECTION

This system enables customers to set up the port over-temperature protection function. Simply click on "Configuration" → "Thermal Protection" to configure and save the settings.

3.2.4 PORTS

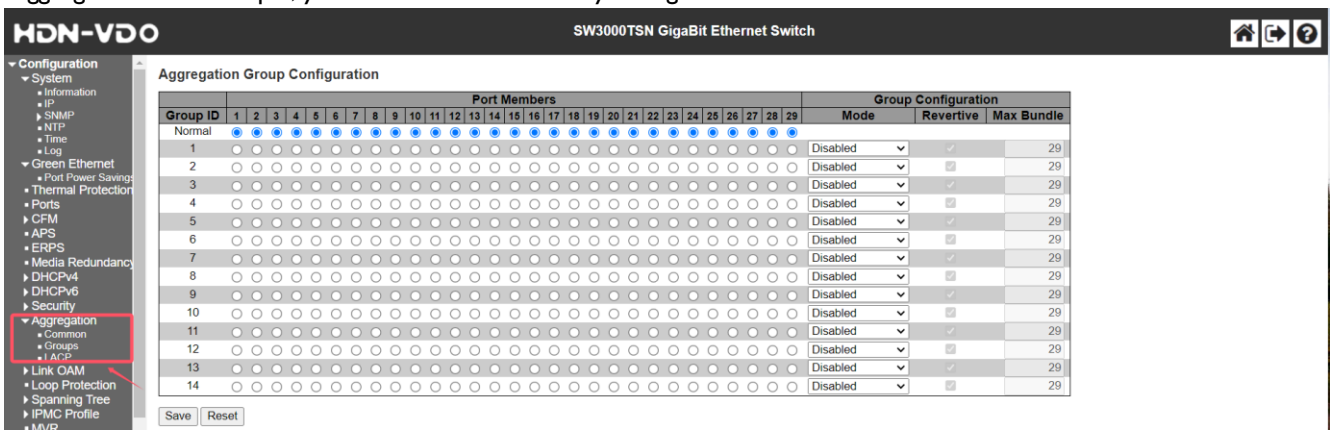
This system allows customers to configure port parameters. Simply click on "Configuration" → "Ports" to make settings and save them. This feature enables the configuration of parameters for each port on the device, including port speed, full/half duplex mode, auto-negotiation, Flow Control, Priority Flow Control (PFC), Maximum Frame Size, Excessive

Collision Mode, Frame Length Check, and FEC Mode (for 2.5/5/10G).



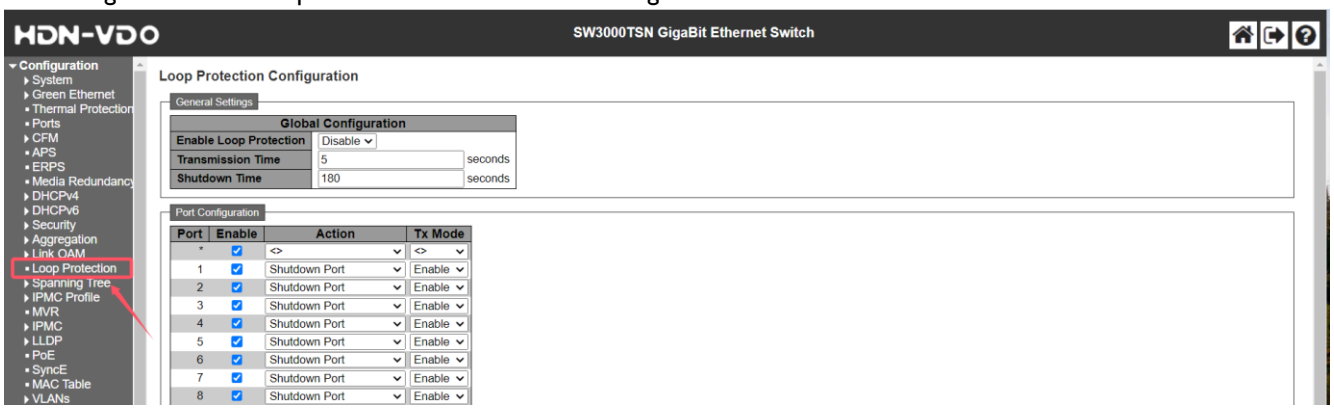
3.2.5 AGGREGATION

This system enables customers to set up customized customer information. By clicking on "Configuration" → "Aggregation" → "Groups", you can make the necessary configurations and save them.



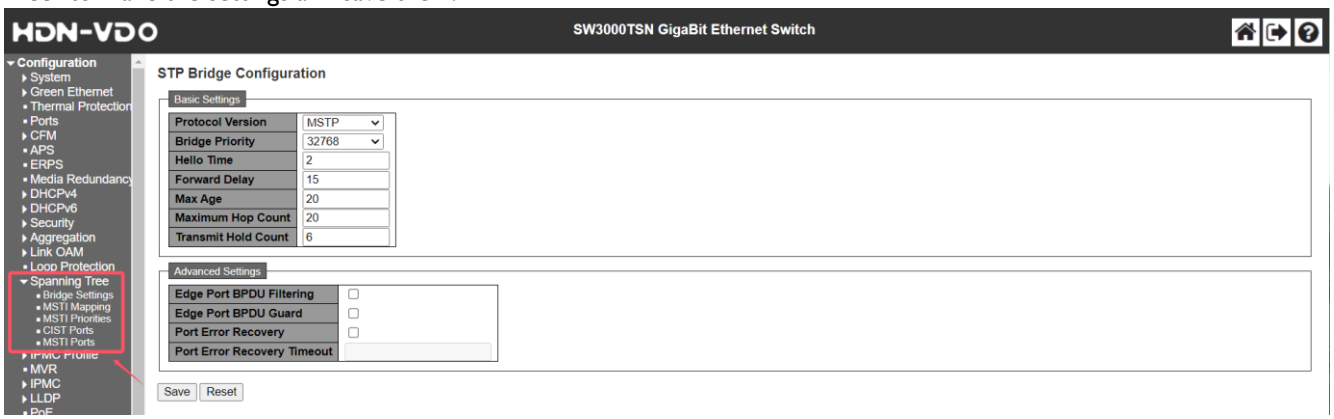
3.2.6 LOOP PROTECTION

This system enables customers to configure loop protection to prevent internal or external loop formation. Simply click on "Configuration" → "Loop Protection" to make the settings and save them.



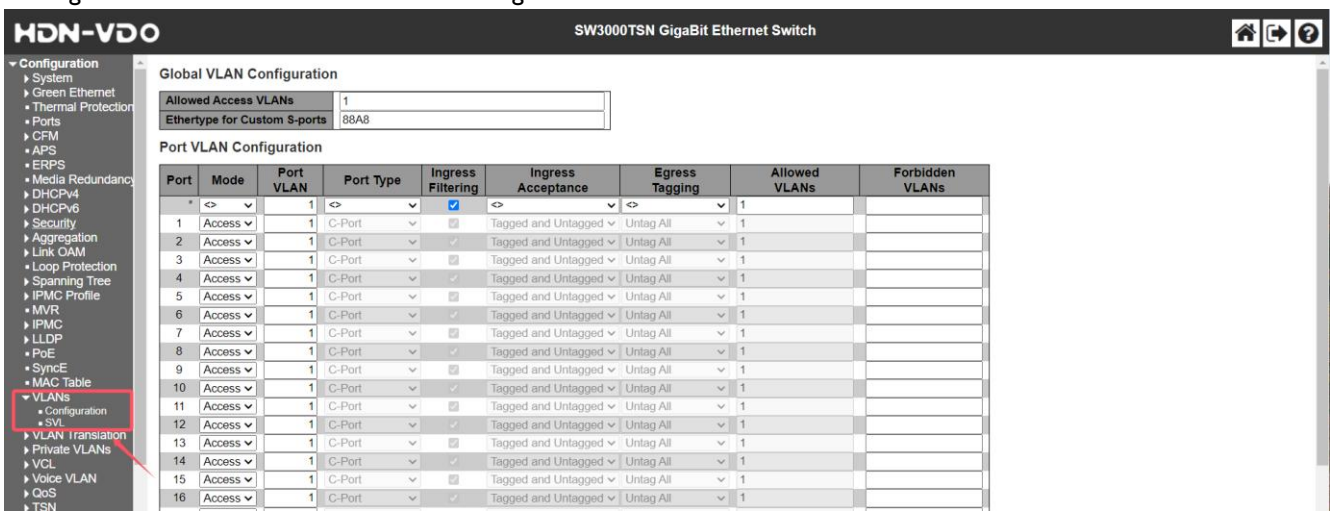
3.2.7 SPANNING TREE

This system allows customers to configure the Spanning Tree protocol. Simply click on "Configuration" → "Spanning Tree" to make the settings and save them.



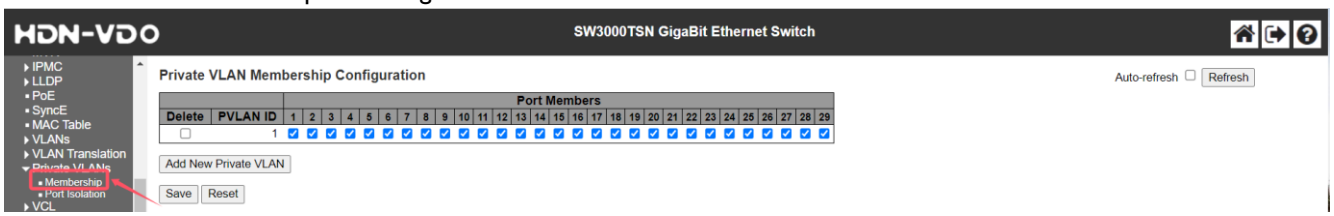
3.2.8 802.1Q VLAN

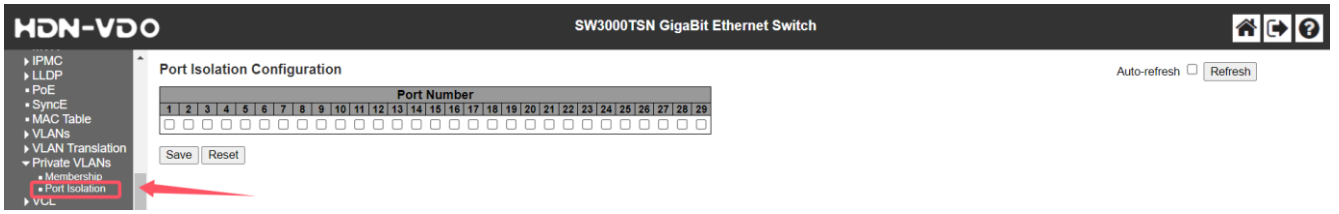
This system enables customers to configure port VLAN IDs and perform internal VLAN segmentation. Simply click on "Configuration" → "VLANs" to make the settings and save them.



3.2.9 ISOLATION AND MEMBERSHIP

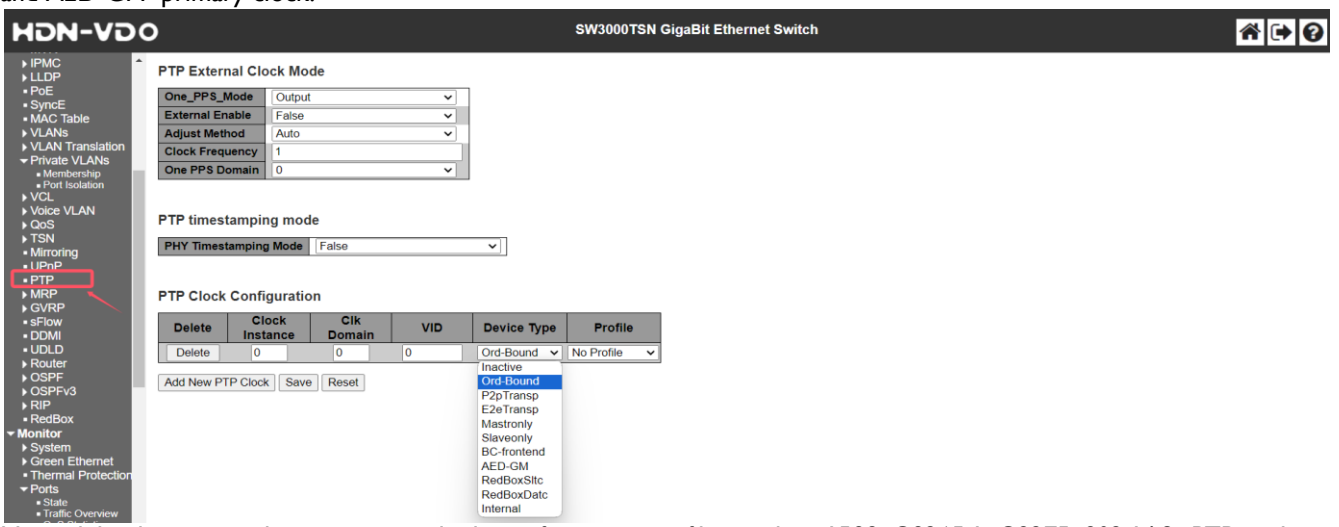
This system allows customers to configure port isolation and port-group communication. Simply click on "Configuration" → "Private VLANs" to set up the configurations and save them.



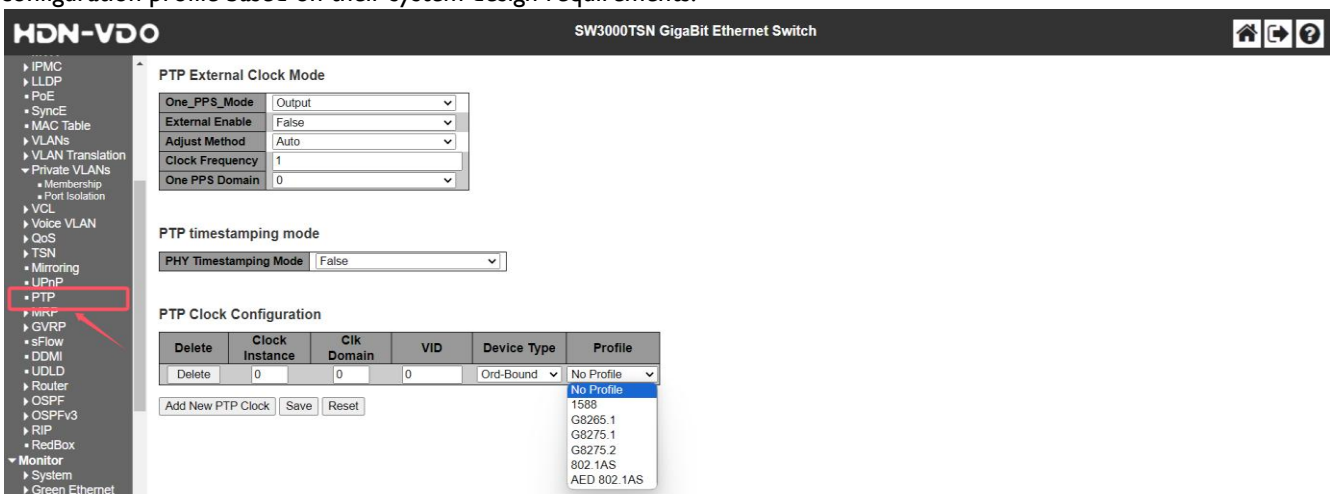


3.2.10 PTP IEEE802.AS,IEEE1588

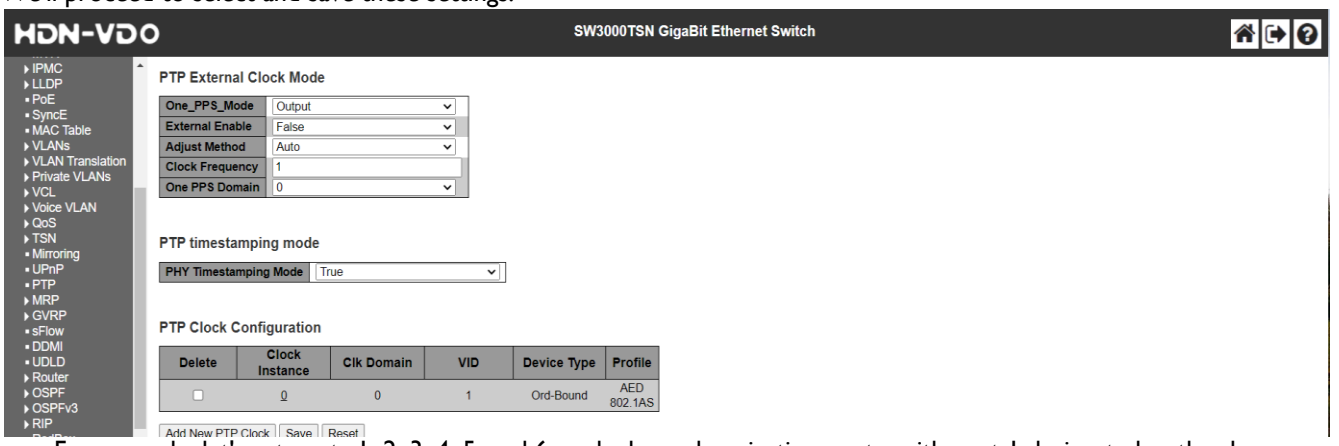
This system supports 802.1as and IEEE 1588 protocols, along with applications such as PTP, gPTP, and gPTP-Automotive. Customers can click on "Configuration" → "PTP" to make settings and save them. As shown below, upon entering the PTP configuration homepage, users can see that the system supports multiple time domains and various types of time synchronization systems, including commonly used options such as BC, P2P Transparent Clock, E2E Transparent Clock, and AED-GM primary clock.



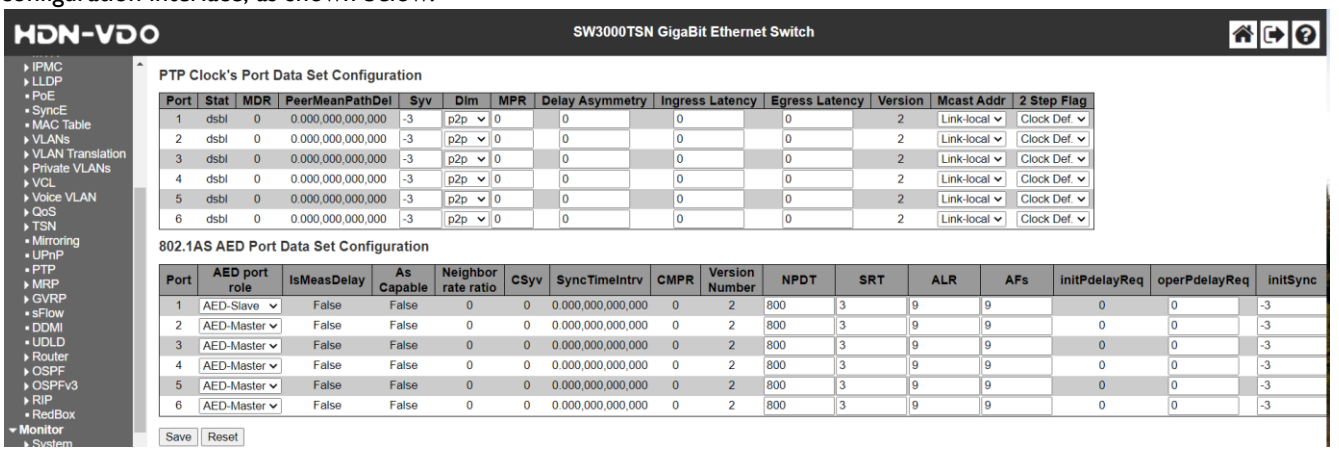
Meanwhile, the system also supports multiple configuration profiles, such as 1588, G8265.1, G8275, 802.1AS gPTP, and AED802.1AS (gPTP-Automotive) used in the automotive sector. Customers can select the appropriate synchronization configuration profile based on their system design requirements.



For instance, let's take selecting the Ord-Bound boundary clock and the AED802.1AS configuration profile as an example. We'll proceed to select and save these settings.

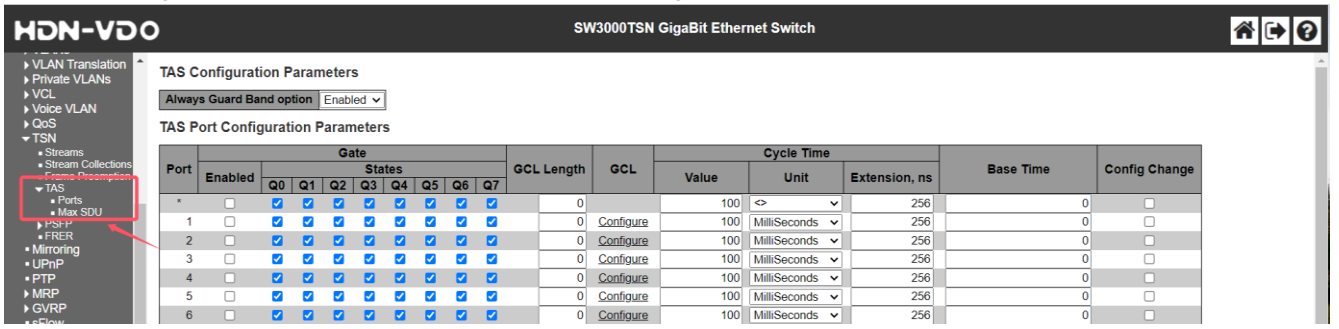


For example, let's set ports 1, 2, 3, 4, 5, and 6 as clock synchronization ports, with port 1 designated as the slave clock (note that a boundary clock can only have one slave clock configured to inherit time from an external master clock, while the other ports function as master clocks). After selecting the ports, you can proceed to the PTP clock parameter configuration interface, as shown below.



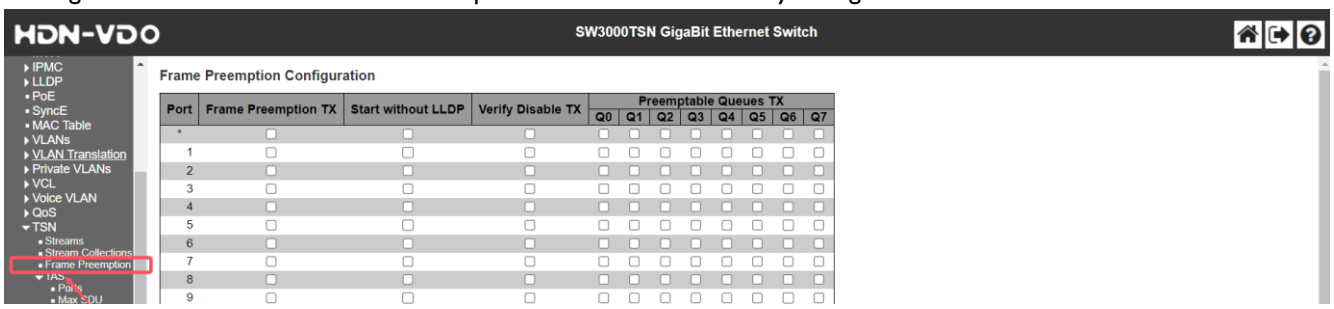
3.2.11 TSN-TAS (802.1QBV)

For the Traffic Shaping (TAS) feature within TSN, the system allows customers to configure TAS settings. Simply click on "Configuration" → "TSN" → "TAS" to make the configurations and save them.



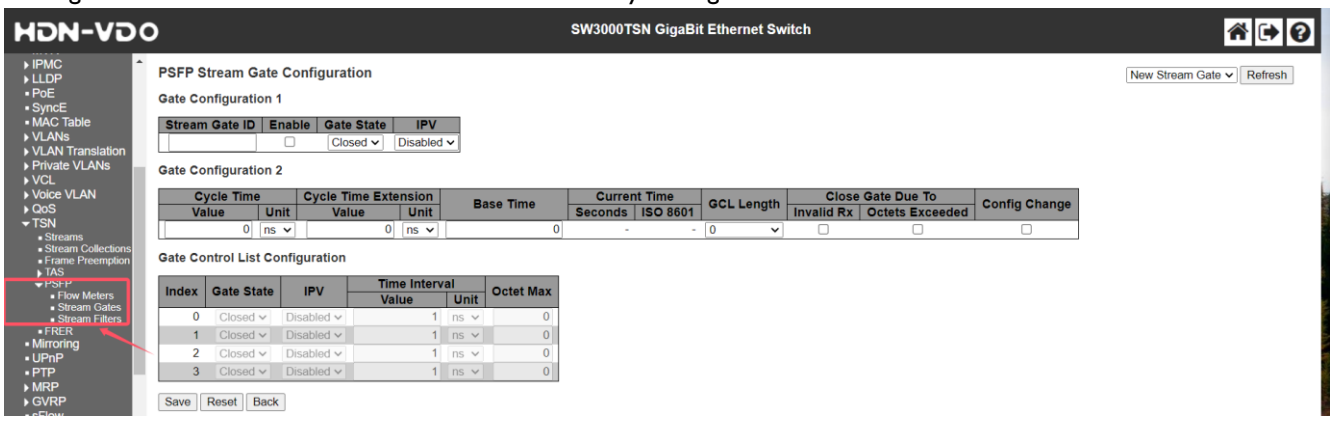
3.2.12 TSN-FRAME PREEMPTION (IEEE 802.1 QBR & IEEE 802.3BU)

For the Frame Preemption feature within TSN, the system enables customers to configure the settings. Simply click on "Configuration" → "TSN" → "Frame Preemption" to make the necessary configurations and save them.



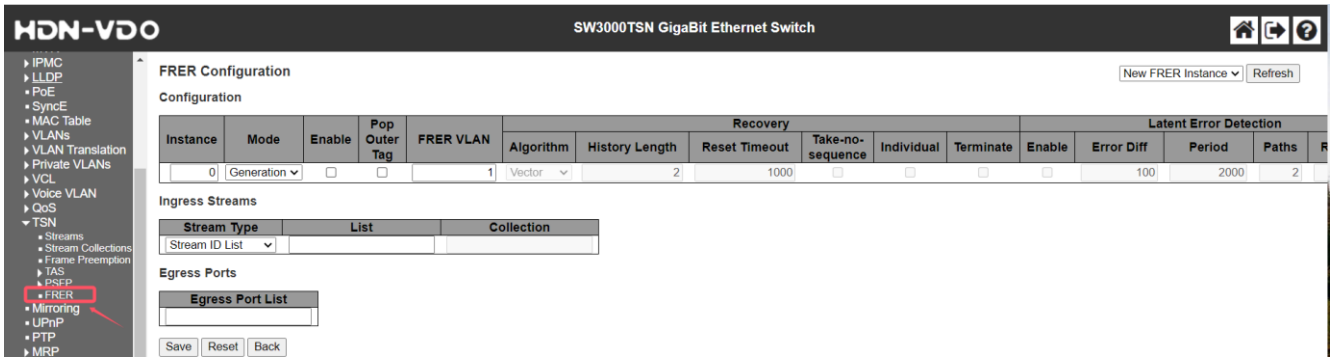
3.2.13 TSN-PSFP(802.1QCI)

The PSFP (Per-Stream Filtering and Policing) feature within TSNallows customers to configure settings. Simply click on "Configuration" → "TSN" → "PSFP" to make the necessary settings and save them.



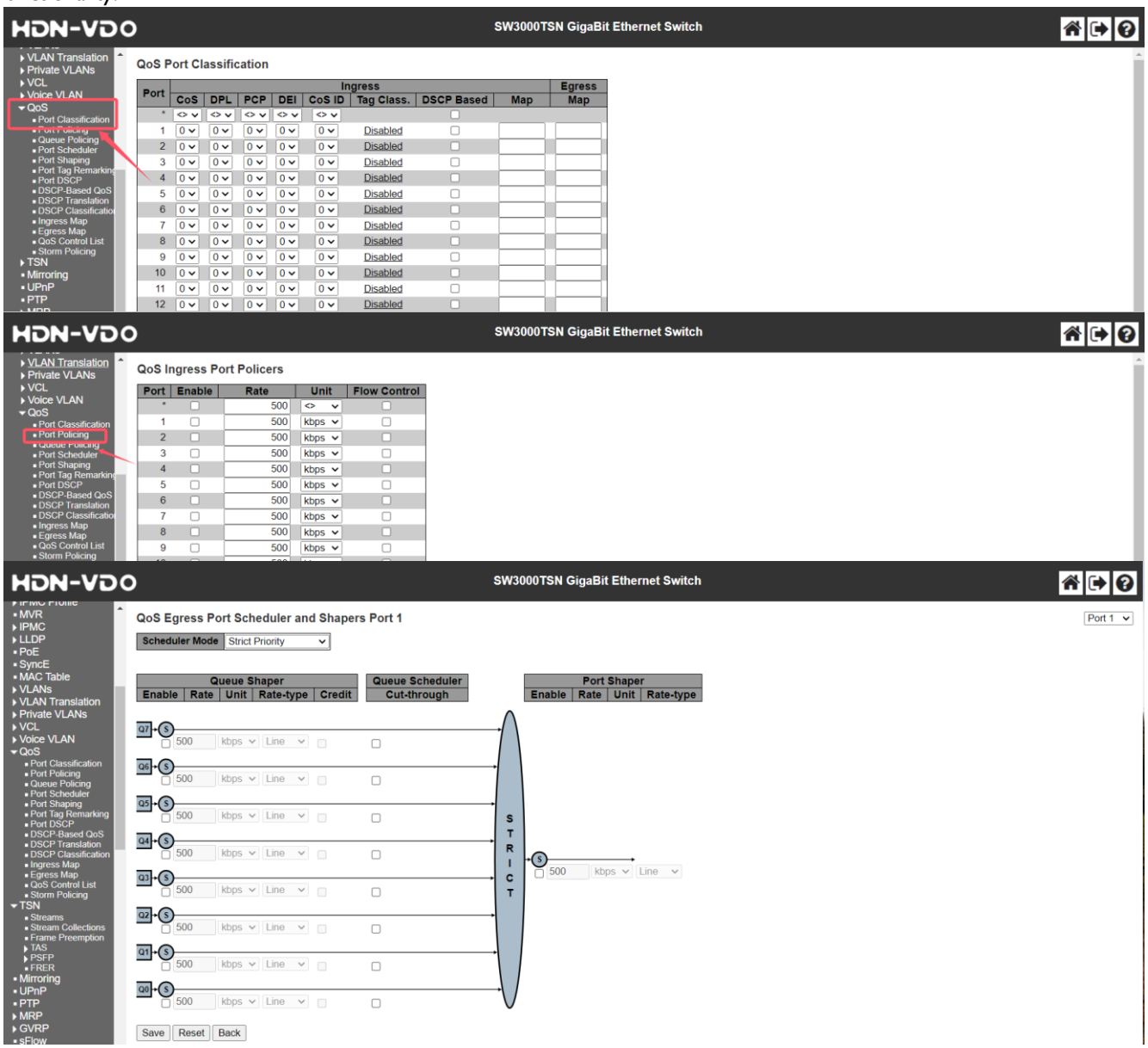
3.2.14 TSN-FRER (802.1CB)

The FRER(Frame Replication and Elimination for Reliability) feature within TSN enables customers to configure redundant backup settings. Simply click on "Configuration" → "TSN" → "FRER" to make the necessary configurations and save them.



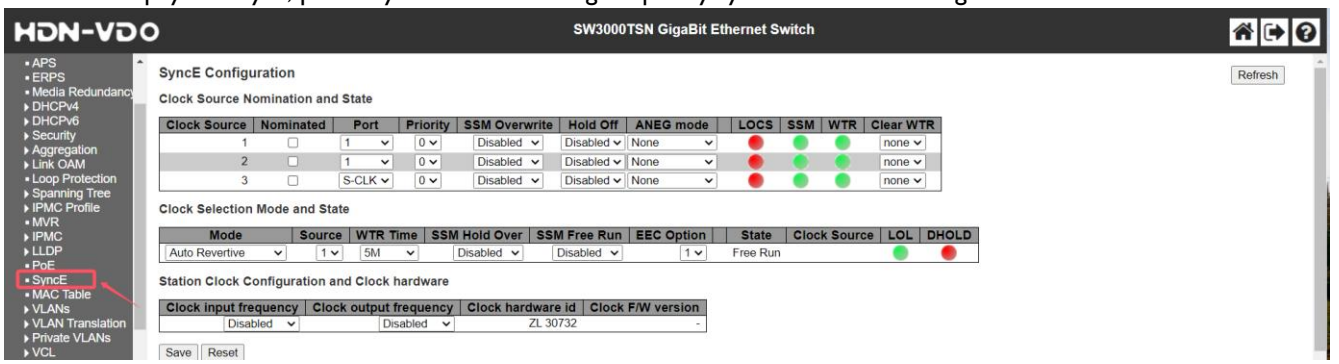
3.2.15 QOS

The system supports Quality of Service (QoS) configuration. Simply click on "Configuration" → "QoS" to make the necessary settings and save them. QoS, or Quality of Service, pertains to the performance metrics of network services. Factors influencing QoS include transmission bandwidth, transmission delay, data packet loss rate, and more. To enhance QoS in a network, measures can be taken such as ensuring adequate transmission bandwidth, reducing transmission delays, minimizing data packet loss, and controlling delay jitter. QoS technologies encompass traffic classification, traffic policing, traffic shaping, rate limiting, congestion management, and congestion avoidance, among others. Furthermore, QoS serves as the foundation for the pre-configuration of frames and streams within TSN (Time-Sensitive Networking) functionality.



3.2.16 SYNCE

The system supports SyncE (Synchronous Ethernet) configuration. Simply click on "Configuration" → "SyncE" to make the necessary settings and save them. Synchronous Ethernet (SyncE) is a clock synchronization technology based on the Ethernet physical layer, primarily used for achieving frequency synchronization among network devices.

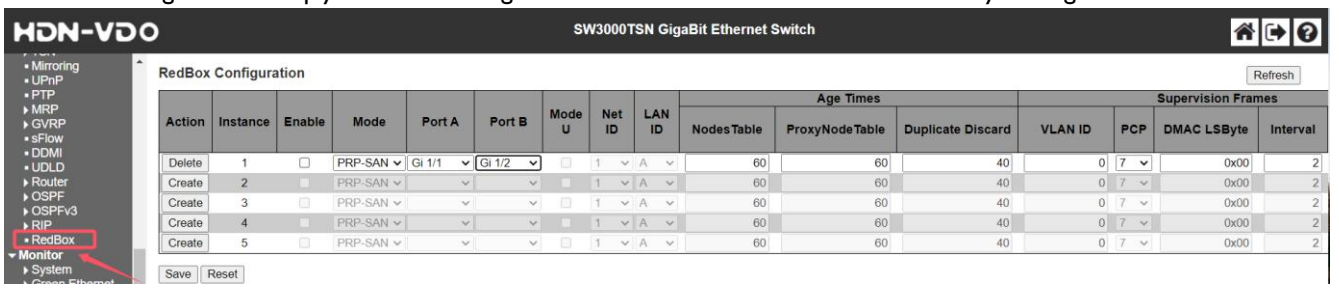


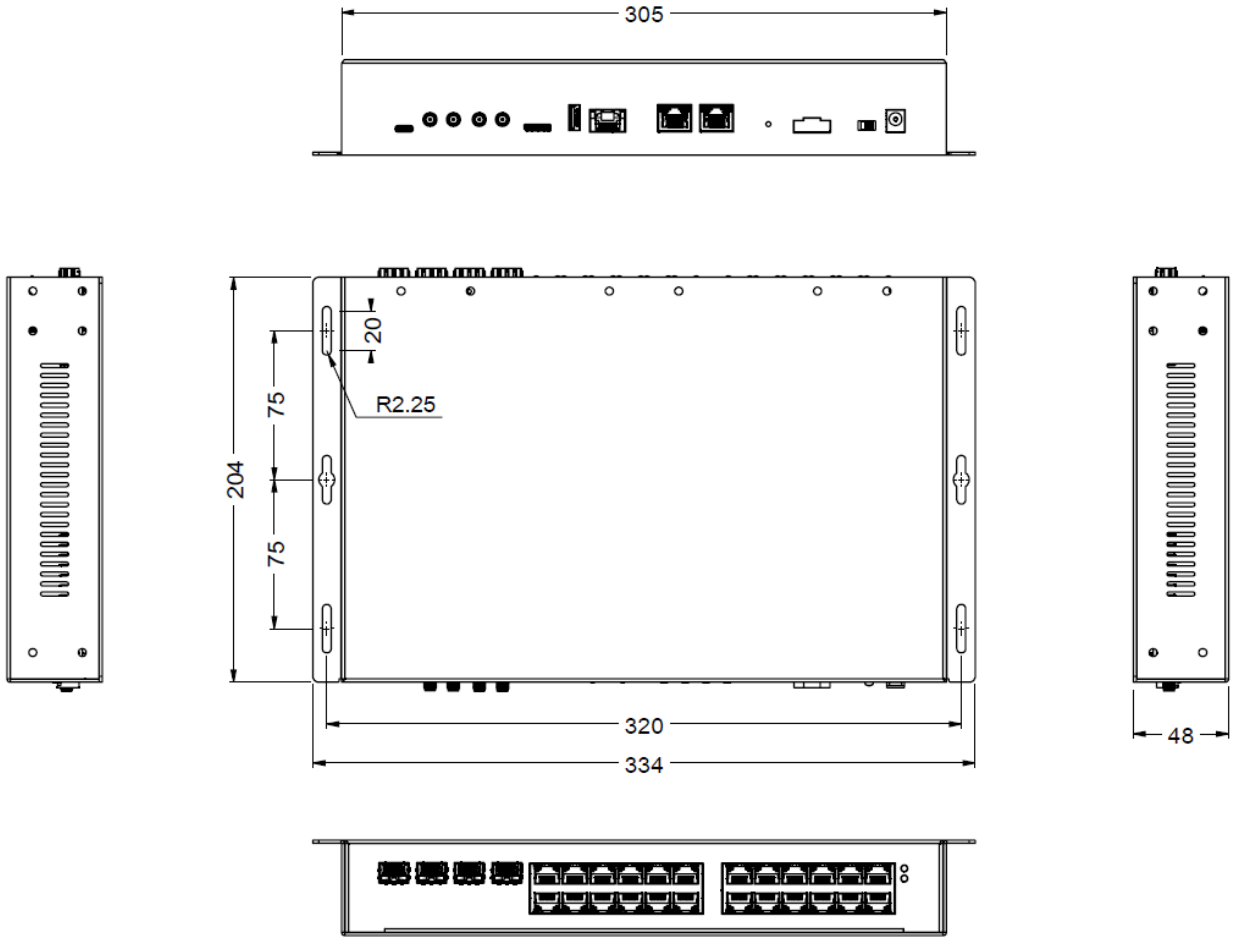
3.2.17 REDBOX

Parts 5 and 4 of the IEC 62349-3 standard define the High-Availability Seamless Redundancy (HSR) protocol and the Parallel Redundancy Protocol (PRP), respectively. Both protocols offer zero-delay recovery times and no frame loss over Ethernet networks. HSR is designed for Ethernet ring topologies, ensuring a known worst-case frame delivery time. PRP, on the other hand, operates with two independent legacy Ethernet networks and is not suitable for real-time scenarios. In fact, PRP, HSR, and PTP can be combined to support time-aware networks. Critical devices requiring fully redundant paths in both real-time and non-real-time sections can benefit from the integration of PRP and HSR networks. PRP consists of two standard Ethernet networks, with PRP-capable devices responsible for sending and receiving duplicate frames over both LAN networks. This approach is well-suited for reusing conventional Ethernet infrastructure but lacks any mechanism to guarantee the worst-case delivery time for a given frame. In contrast, HSR allows for the calculation of this parameter, compensating for its limitation. Therefore, it represents a standardized and interoperable solution for control-oriented communications.

For example, a typical zero-delay recovery time topology for IEC 61850 substation/aircraft applications is based on implementing PRP in the station bus section and an HSR ring in the process bus area. The connection between the PRP network and the HSR ring must be made through two distinct points to avoid the aforementioned SPOF (Single Point of Failure, a critical component in a system whose failure would cause the entire system to collapse). The IEC 62349-3 standard defines the behavior of devices involved in this interconnection. Specifically, when Ethernet frames traverse from one network to another, they need to support the HSR/PRP mode to properly manage the sequence number field. From an integrator's perspective, one option is to combine and configure standalone RedBox devices that support PRP-HSR and can handle (combined) redundant IEEE 1588 profiles.

The system's RedBox functionality enables a seamless merging solution for HSR/PRP networks and supports customer configuration. Simply click on "Configuration" → "RedBox" to make the necessary settings and save them.



Attach:

SW3000R TSN Size